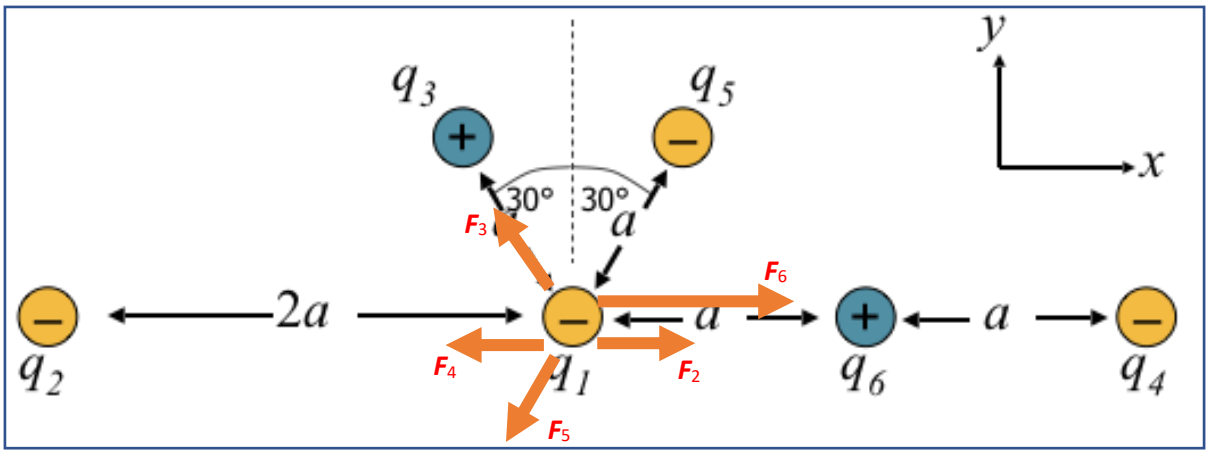
Practice Exam 2023SUM – Gradescope Submission

POTENTIAL SOLUTION SET

Historical Note: *this practice exam is an adaptation of the 2019 Physics 2:PST exam.*

Q1. a) In your script book, draw and label all the forces acting on q_1, including relative direction and magnitude.



b) Find a functional expression for the net force on q_1 due to the charges in terms of q, a and θ.

$$F_1 = F_2 + F_3 + F_4 + F_5 + F_6$$

From above, list F_j & directions. $F_2: +x;\ F_4: -x;\ F_6: +x;\ F_{3,5}: +, -y;\ +x$ out of line at $\theta = 30°$

Only x components remain → $F_1 = \frac{kq^2}{a^2} - \frac{2kq^2}{a^2}\sin\theta$

c) Evaluate the expression in (b) to find the value of the net force and its direction on q_1.

0 N (no direction)

d) If q_1 is displaced towards the right along the line of charges, give a formula for the net force and direction. [Hint: displace it by δx in the x direction and take derivatives.] Is the charge q_1 trapped?

$$F_x = k\frac{q^2}{(2a)^2} - k\frac{q^2}{(2a)^2} + k\frac{q^2}{a^2} - k\frac{q^2}{a^2}\sin 30^\circ - k\frac{q^2}{a^2}\sin 30^\circ$$

$$= 0$$

$$F_y = k\frac{q^2}{a^2}\cos 30^\circ - k\frac{q^2}{a^2}\cos 30^\circ$$

$$F_x' = k\frac{q^2}{(2a+\delta x)^2} - k\frac{q^2}{(2a-\delta x)^2} + k\frac{q^2}{(a-\delta x)^2} - \left\{k\frac{q^2}{\left(a\frac{\sqrt{3}}{2}\right)^2+\left(\frac{a}{2}+\delta x\right)^2}\right\}\sin\theta_3' + k\frac{q^2}{\left(a\frac{\sqrt{3}}{2}\right)^2+\left(\frac{a}{2}-\delta x\right)^2}\sin\theta_5'$$

$$F_y' = \{\ \}\cos\theta_3' - \{\ \}\cos\theta_5'$$

$$F_x' = \frac{\partial F}{\partial x}\delta x = kq^2\delta x\left\{\frac{-2}{(2a+\delta x)^3} - \frac{-2}{(2a-\delta x)^3} + \frac{-2}{(a-\delta x)^3} \ldots\right\}$$

Key points to note:

- x-component – repelled by Hooke's law so acts as a **trap in the x-direction**.
- y-component – if it were three symmetric positive or negative charges it would also trap in the y-direction but in this case q_5 dominates and the force is in the **-y direction**

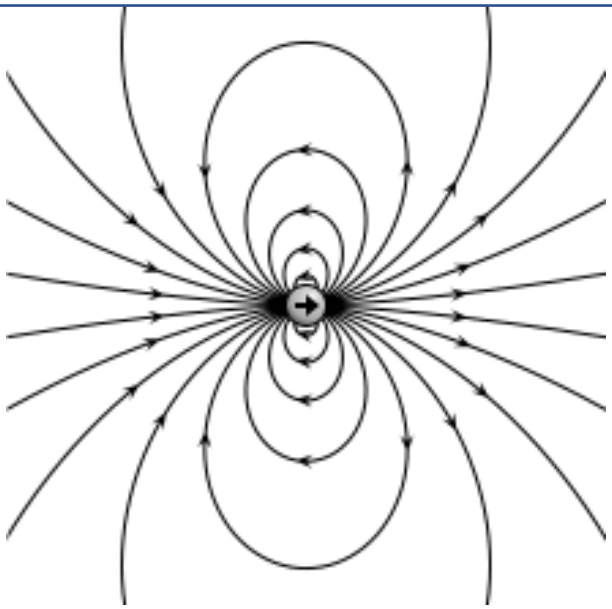
e) Explain why Coulomb's Law obeys $1/r^2$ using a diagram, with reference to concepts of flux and electric field. State the Principle of Superposition and explain why the electrical force obeys the Principle of Superposition. Explain why the electrical force obeys vector addition.

The electric flux through a unit area represents the strength of the electric field and force. Because we live in a universe with three spatial dimensions, the surface area through which the flux travels is 4 pi r^2 .Therefore the flux per unit area, the field strength and ergo the force goes as $1/r^2$ from a point source.

Principle of Superposition: forces add, vector fields add, scalar fields add. The electric force obeys the Principle of Superposition because the force is linear with charge. Any linear function will add linearly.

The electrical force obeys vector addition because the functional is linear [Principle of Superposition] and because the field is a vector [so it adds as a vector].

f) i. Draw the electric field lines of an electric dipole.



ii. State the rules you are using in drawing the field lines.

Originate on positive charge [or at infinity].

Terminate on negative charge [or at infinity].

All lines continuous except at source or sink.

Strength of field and force on a charge proportional to the flux [number of field lines] per unit cross-sectional area.

Q2. a) A point positive charge, +Q, is placed at the centre of a spherical cavity of radius R formed in a large block of conducting material. Use Gauss' Law to find the magnitude and direction of the electric field at a distance:

i) r = R/2, inside the cavity;

By symmetry, all electric field vectors will point radially outwards. Draw a spherical Gaussian surface at r = R/2.

$$\int \mathbf{E}.\mathrm{d}A = \frac{q_{\text{enc}}}{\epsilon_0}$$

$$\rightsquigarrow E \times (\text{surface area of sphere}) = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$\rightsquigarrow E = \frac{Q}{4\pi r^2 \epsilon_0}$$

Substitute $r = R/2$:

$$\rightsquigarrow E = \frac{Q}{\pi R^2 \epsilon_0}$$

ii) $r = R$, at the surface of the cavity;

Same analysis as above, but substitute $r = R$:

$$E = \frac{Q}{4\pi R^2 \epsilon_0}$$

and the field continues to point **radially outwards**.

iii) r > R from the centre of the cavity, inside the conductor.

$r > R$ Because the material is conducting, a screening charge forms on the inner surface of the cavity. Thus, any Gaussian surface will have $q_{\text{enc}} = Q + \text{screening charge} = Q + (-Q) = 0$. Thus $\int \mathbf{E}.\mathrm{d}A = 0$. Since we have spherical symmetry, $E = 0$.

b) Two long concentric conducting cylinders of radius a and b, respectively, each carry equal and opposite charges, with the inner shell having a linear charge density $+\lambda$ C m^{-1}. The cylinders are of negligible thickness.

Assuming that $b > a$, use Gauss's law:

i) to show that $\boldsymbol{E}$ vanishes for $r < a$ and $r > b$;

Consider a cylindrical Gaussian surface of length L and radius r.
For $r < a$, $q_{\text{enc}} = 0$.
For $r > b$, $q_{\text{enc}} = q_a + q_b = L\lambda + (-L\lambda) =$.
Thus:

$$\int \mathbf{E}.\mathrm{d}\mathbf{A} = 0$$

Symmetry dictates the ends of the cylinder do not contribute.

$$\rightsquigarrow E \times L \times 2\pi r = 0, \qquad L \neq 0, r \neq 0$$

$$\rightsquigarrow E = 0.$$

ii) to show that for $a < r < b$, $|\boldsymbol{E}| = \frac{\lambda}{2\pi r \varepsilon_0}$;

Same shaped Gaussian surface.

$$\int \mathbf{E}.\mathrm{d}A = \frac{q_{\text{enc}}}{\epsilon_0} = \frac{\lambda L}{\epsilon_0}$$
$$\rightsquigarrow |E| L 2\pi r = \frac{\lambda L}{\epsilon_0}$$
$$\rightsquigarrow |E| = \frac{\lambda}{2\pi r \epsilon_0}.$$

and iii) to describe the direction of the electric field in the space between the cylinders.

Consider a cylindrical Gaussian surface, a < r < b. As L ⟶ 0, q_{enc} ⟶ 0 and so the electric field must have only radial components.

Q3. a) Professor Cuthbert Calculus is an absent-minded scientist. You visit him at his laboratory. He invites you to assist him in the following experiments.

i) He has a large conducting rod on a wall, pointing out from the wall, which he has charged up to 10 000 V. You can hear some fizzing and crackling coming from the tip. He switches off the machine and asks you to ground the rod. Assuming that you agree, explain how you would do so safely.

It is safe with a rod of metal to connect from the rod to the ground, if it is insulated from you by e.g. a wooden handle of some useful length. 'Ensure that you do not form part of the circuit.'

ii) He asks you to keep your hands on the surface of a Van de Graaff accelerator which will be slowly raised to a maximum potential of 100 kV. The sphere on the Van de Graaff is 0.5 m off the floor. The initial current to the sphere is about 100 uA and the sphere accumulates 10 uC at the maximum potential over a period of a few seconds. Air breaks down (ionizes) at E = 3 × 10^6 N/C. Assume any discharge occurs over a period of 1 ms.

The bottom of your shoes has a thickness of 3 mm. Explain what you would expect to occur [Give potential differences and currents in your answer.]

Charge will build up until the potential reaches 3mm x 3M N/C = 9000 V at which it will arc from your foot to ground, in perhaps 0.2ms, giving a current of I = Q/t = 5 or so μC / 1 ms = 5 mA. In this case, you would feel regular arcs, although they would not be dangerous. [it could be worse if it were e.g. 8 mm thick.]

b) Nikola Tesla is pictured below, with discharges of several million volts cascading around him in his Colorado Springs Laboratory. The roar that accompanied such discharges could be heard ten miles away. Tesla is unharmed. Why? Explain in terms of electric field lines and conductors.

No current passes through him – he does not complete a circuit. He is in a Faraday cage, i.e. he is inside a conducting surface, so no electric fields are inside.

Q4. a) Using the Biot-Savart Law, show that the magnetic field due to an infinitely long straight wire carrying a current *i* amperes, at a radial distance *r*, has magnitude given by $B = \frac{\mu_0 i}{2\pi r}$ and explain direction.
[Hint: you might find the identity $\int \frac{ydx}{[x^2+y^2]^{3/2}} = \frac{x}{y[x^2+y^2]^{1/2}}$ useful.]

$$d\boldsymbol{B} = \frac{\mu_0}{4\pi}\frac{id\boldsymbol{s}\times\boldsymbol{r}}{r^3}$$

Integrate Biot-Savart along wire.

$$\int_{-\infty}^{\infty}\frac{Rds}{[s^2+R^2]^{3/2}} = 2\int_0^{\infty}\frac{Rds}{[s^2+R^2]^{3/2}} = 2\left[\frac{s}{R[s^2+R^2]^{\frac{1}{2}}}\right]_{s=\infty} = \frac{2}{R}$$

So $\int d\boldsymbol{B} = \int \frac{\mu_0}{4\pi}\frac{id\boldsymbol{s}\times\boldsymbol{r}}{r^3} = \frac{2\mu_0 i}{4\pi r} d\hat{\boldsymbol{s}}\times\hat{\boldsymbol{r}}$

$B = \frac{\mu_0 i}{2\pi r}$ Determines magnitude. Direction = RHR from cross product

b) Show that your answer to part a) above is consistent with Ampere's Law.

$\int \boldsymbol{B}\cdot d\boldsymbol{s} = \mu_0 i = B\ 2\pi r$. loop around wire in circle. Determines magnitude. Direction = RHR

c) Imagine that we have a current-carrying cylinder made from a perfect conductor, with a cylindrical hole through its centre. The inner radius is *a* m and the outer radius is *b* m. The current direction is perpendicular to the page in the diagram below. The cross-sectional current density in the cylinder is *J* Am^{-2}. Indicating any assumptions you make, determine the magnetic field $\vec{B}$ inside the cylinder for *a<r<b*.

$$i = \pi J(b^2 - a^2)$$

inside Amperian loop of radius r, $i = \pi J(r^2 - a^2)$

$\longrightarrow B(r) = \frac{1}{2}\mu_0 J(r - \frac{a^2}{r})$

Same direction.

Q5.

(a) A particle of mass m and charge q is moving in a circular path in a magnetic field B. Show that its kinetic energy is proportional to r^2, the square of the radius of its path. [Hint: use the equation for centripetal motion, F = mv^2/r]

The magnitude of the force from the magnetic field is

$$F = qvB$$

and points perpendicularly towards the centre of the circular path. As there are no other forces, then this must be centripetal force and thus

$$qvB = \frac{mv^2}{r}.$$

Solving the above equation for v gives

$$v = \frac{qB}{m}r$$

and thus the kinetic energy is

$$E_K = \frac{1}{2}mv^2 = \frac{(qB)^2}{2m}r^2$$

If the student uses $F = iLB$ and correctly equate it to the centripetal force,

$$iLB = \frac{mv^2}{r}$$
$$v^2 = \frac{iLBr}{m} = \frac{2\pi iBr^2}{m}$$
$$E_K = \frac{1}{2}mv^2 = \pi iBr^2$$

(b) (i) A demonstration showed a fan-shaped plate of aluminium swinging through a large 'C' magnet, slowing down on each pass through the magnet and then stopping. Why did the aluminium plate slow down as it entered the magnetic field? Use the RH grip rule in your answer to explain the sign of the effect.

The formula for induced emf in response to changing magnetic flux (Faraday's law) is EMF = $-N\frac{\Delta\Phi_B}{\Delta t}$, $\Phi_B = \Sigma B A \cos\theta$. In this case the field is varying on any region of the plate surface. Here N=1.

Because it is a conductor the EMF drives a circuit EMF = *IR* where *R* is some effective resistance of the aluminium. Hence a current loop forms and following the RH Grip rule produces a field opposing the original change of flux $\sum_{loop} B.\Delta l = \mu I$.

By Lenz's law these oppose the change in magnetic flux as the magnet falls through. Hence the induced fields oppose the motion and slow the magnet down dramatically. $F = IlB$ sin theta (for the side l in the field and perpendicular to the motion of the plate).

The motion is therefore damped by induction (eddy currents) induced in the plate.

(ii) A different plate now replaces that in (i), with many slots in the new plate along the axis of rotation, separated by 2 mm, and the behaviour of the swinging fan changes dramatically. Compare the number of oscillations of this plate to the behaviour of the plate in (i). Explain in terms of your answer to part (a), eddy currents and the applicable formula describing their cause.

The plate's speed is much less damped. The emf induced is reduced, because the radius of the loops (i.e. the maximum area A) is restricted.

The cuts in the plate are open circuits for any large current loops that might otherwise be formed. Now the electrons cannot go around in a large circle in the plane of the plate (i.e. perpendicular to the field change). Less energy dissipated as current/electron movement.

Q6. **(a)** The mass of a nitrogen molecule (N2) is 28u.
(i) Compute the rms speed of a nitrogen molecule at 20.0°C.
At what temperatures will the rms speed be
(ii) half that value and
(iii) twice that value?

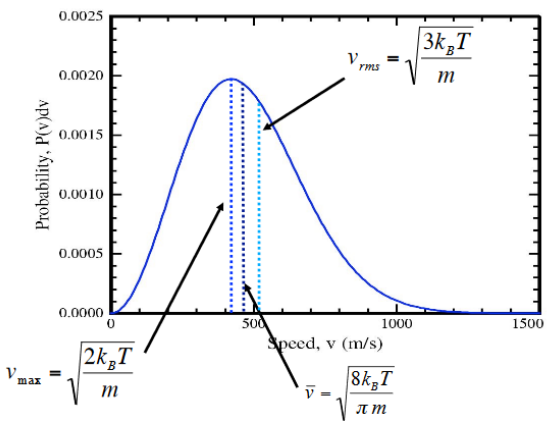
(i) We have $v_{rms} = \sqrt{\frac{3k_BT}{m}} = \sqrt{\frac{3 \cdot k_B \cdot 293}{28u}} = 511m/s$

(ii) We need $v_{rms} = \sqrt{\frac{3k_BT}{m}} = \frac{1}{2}\sqrt{\frac{2k_B293}{m}}$, which simplifies to give $T = \frac{293}{4} = 73K$

(iii) Likewise, we have $v_{rms} = \sqrt{\frac{3k_BT}{m}} = 2\sqrt{\frac{2k_B293}{m}}$, so $T = 293 \cdot 4 = 1172K$

(b) In the kinetic theory of gases explain what is meant by the "Maxwell speed distribution".

- Particles in a gas don't have one defined speed, but are more likely to have speeds in a particular range dependent on their temperature
- The Maxwell speed distribution plots these possible speeds for a particular gas at a particular temperature, and allows us to determine the max speed, the average speed, and the root-mean-square speed.



Q7.

a) In a Compton collision with an electron inside rubidium, a photon of violet light (λ = 4000Å) is backscattered through an angle of 180 ∘ .

i) How much energy (in eV) is transferred to the electron in this collision?

If angular change = 180 degrees, Compton shift ~0 nm ∴ $\lambda_{new} = 400nm$

Momentum change by incident photon $p = 2\times\frac{h}{\lambda} = 3.31\times10^{-27}kgms^{-1}$

$$\rightarrow |\Delta p_\gamma| = |\Delta p_e|$$

$\rightarrow E_e = \frac{p^2}{2m_e} = 6.03\times10^{-24}J = \mathbf{0.0376\ meV}$

ii) Could violet light eject electrons from the rubidium? Explain.

Yes (most likely) – as this wavelength is shorter than 550nm, which was shown to eject electrons in the related short-answer (Canvas quiz) question. Some calculatory argument:

$$KE = hf - E_0$$
$$= \frac{hc}{\lambda} - E_0$$
$$= \frac{6.63\times10^{-34}\times3\times10^{8}}{400\times10^{-9}} - (3.46\times10^{-19}) = 1.51\times10^{-19}J = 0.944eV$$

∴There is sufficient energy in the violet photons to induce release of photoelectric electrons from the Rubidium.

7b) An electron and a proton are accelerated to 25.0% the speed of light. What are the wavelengths associated with each particle in nm?

(m_e = 9.11 x 10^{-31}kg; m_p = 1.67 x 10^{-27} kg; neglect relativistic effects.)

$$v_p = v_e = 0.25c \rightarrow \lambda_e, \lambda_p ? \rightarrow p = mv \equiv \frac{h}{\lambda}$$
$$p_e = (9.11\times10^{-31}\times0.25\times3\times10^{8}) = 6.83\times10^{-23}kgms^{-1}$$
$$p_p = (1.67\times10^{-27}\times0.25\times3\times10^{8}) = 1.25\times10^{-19}kgms^{-1}$$
$$\therefore \lambda_e = \frac{h}{p_e} = 9.7\times10^{-12}\text{m} = \mathbf{9.7\times10^{-3}nm}$$
$$\lambda_p = \frac{h}{p_p} = 5.29\times10^{-15}\text{m} = \mathbf{5.29\times10^{-6}nm}$$

7c) An electron moves with a speed of 1.5 x 10^7 m/s, with uncertainty on its speed of 1 %.

i) What is the absolute value on the uncertainty of the electron's speed? If the location of the electron is simultaneously measured, what is the minimum uncertainty (Δx) in knowing the electron's location?

$$v = 1.5\times10^{7}\ \text{ms}^{-1} \rightarrow \Delta v = 1.5\times10^{\mathbf{5}}\ \text{ms}^{-1}$$
$$p = mv = 9.11\times10^{-31}\times1.5\times10^{7} = 1.37\times10^{-23}\ kgms^{-1}$$
$$\frac{\Delta v}{v} = 0.01 = \frac{\Delta p}{p} \therefore \Delta p = 1.37\times10^{-25} kgms^{-1}$$
$$\rightarrow \Delta x\Delta p \geq \frac{\hbar}{2} \rightarrow \Delta x \geq \frac{\hbar}{2\Delta p} = \mathbf{3.86\times10^{-10}m}$$

ii) Electrons have a diameter of about 10^{-17} m. Compare this with □x, what can you conclude?

Can't know anything (features, dimensions, etc) smaller than 10^{-10}m, to determine if the electron diameter is really 10^{-17}m → would need a more "inaccurate"/uncertain momentum value *p* for a better-defined *x*.

Q8. a) In classical physics an atom would not be stable because it would emit electromagnetic waves of frequency *f*, because the electron's orbit, seen edge on, can be seen as an electric dipole. At what electron radius, in nm, would the electron emit light with a wavelength of 600 nm?

Photon frequency $f = \frac{c}{\lambda} = 5\times10^{14}$ Hz or photon energy $E = \frac{hc}{\lambda} = 3.31\times10^{19}$ J

Photon frequency = frequency of electron orbit

Centripetal force $F = \frac{m_e v^2}{r} = \frac{1}{4\pi\epsilon_0}\frac{Ze^2}{r^2}$

Electron velocity $v = \sqrt{\frac{1}{4\pi\epsilon_0}\frac{Ze^2}{m_e r}}$

Electron orbit frequency $f = \frac{1}{T} = \frac{v}{2\pi r} = \sqrt{\frac{1}{16\pi^3\epsilon_0}\frac{Ze^2}{m_e r^3}}$

Electron radius $r = \left(\frac{Ze^2}{16\pi^3\epsilon_0 m_e f^2}\right)^{\frac{1}{3}}$

Assume hydrogen $Z = 1$

$$r = \left(\frac{(1.6\times10^{-19})^2}{16\pi^3\times8.85\times10^{-12}\times9.11\times10^{-31}\times(5\times10^{14})^2}\right)^{\frac{1}{3}} = 2.95\times10^{-10}\ \text{m} = 2.95\ \text{A}$$

b) Why in an atom are the energies of the electrons negative?

A free electron (infinitely far away from the nucleus) is defined as having zero potential energy. It is energetically favourable for an electron to move closer to the nucleus and become bound to form an atom as it reduces potential energy. Hence, the bound states of an electron in an atom are negative.

c) A hydrogen atom makes a transition from the n = 2 to the n = 1 energy level. Find the wavelength and frequency of the emitted photon.

Frequency of emitted photon $f = cR\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = 2.46\times10^{15}$ Hz

Wavelength $\lambda = \frac{c}{f} = 122$ nm

d) In interstellar space, highly excited hydrogen atoms, called Rydberg atoms, have been observed. These can be described by standard Bohr atom theory.

i) Find the wavelength radio astronomers must tune to, so they can detect signals from electrons dropping from the n = 273 level to the n = 270 level.

Energy $E = 13.6\ eV\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = 10.2\ eV$ or frequency $f = \frac{E}{h} = 9.8\times10^8$ Hz

Wavelength $\lambda = \frac{hc}{E} = 0.305$ m

ii) What is the radius of the Rydberg atom for which n = 273?

$r = a_B n^2$

$r = 52.92\times10^{-12}\times273^2 = 3.94\ \mu$m

iii) What is the speed of an electron in this Rydberg atom?

$v_n = \frac{c\alpha_{em}}{n} = 8021$ ms^{-1}

iv) Compare the results in ii) and iii) with the radius and velocity of hydrogen atoms in their lowest state. What did you learn from this comparison?

For $Z = 1, n = 1$:

$r = a_B = 52.92\times10^{-12}$ m or numerical comparison to r_{273}

$v = c\alpha_{em} = 2.19\times10^6$ ms^{-1} or numerical comparison to v_{273}

Statement about the Rydberg atom having a much larger radius and a much slower electron than the ground state Hydrogen atom

Q9.

(a) The Molybdenum is transported to hospitals and produces the metastable Technetium to be injected into the patient.

(i) Mo-99 decays to the metastable Tc-99m via beta decay. Write the decay equation and justify whether the decay is β- or β+.

$^{99}_{42}Mo \rightarrow {}^{99m}_{43}Tc + {}^{0}_{-1}\beta + \bar{\nu}$ Beta-minus decay illustrated by A & Z numbers.

(ii) Show that the energy released during this decay is 1.216 MeV.

(Atomic mass of Mo) = 98.9077119u (Atomic Mass of TcM) = 98.9064065u

$$\rightarrow m_{Mo} = 98.9077119\text{u}\times931.49\frac{\text{MeV}}{\text{u}} = 92.131\text{ GeV}$$

$$m_{Tc(m)} = 98.9064065\text{u}\times931.49\frac{\text{MeV}}{\text{u}} = 92.1303286\text{ GeV}$$

$$\Delta E = (92.131545 - 92.1303286)\times10^3\text{ MeV} = \mathbf{1.216\ MeV}$$

Note: Atomic masses *include* electron masses, so the mass of the beta particle is included within the value of the Tc(m) and does not need to be counted again.

(b) To diagnose a patient, they are injected with the metastable Technetium which has been separated from the Molybdenum. The resulting gamma rays emitted as the isotope decays is an excellent radioactive source for imaging.

(i) Calculate the percentage of Tc-99m that will still be present in the patient's body after 1 day.

$$N = N_0 e^{-\left(\frac{\log_e 2\times t}{\tau_{0.5}}\right)} \rightarrow \frac{N}{N_0}\times100\% = e^{\left(\frac{24hours}{6.0058hours}\right)}\times100 = \mathbf{6.3\%}$$

(ii) Both Mo-99 and Tc-99 are beta sources, yet only Tc-99 remains in the patient's body as Mo-99 is removed from the sample before the procedure. Using the data in the table, compare the amount of radiation received from each of the sources and comment on why it is fine for Tc-99 to remain in the body, but not Mo-99. Include calculations to justify your answer.

Compare the half lives of the relevant atoms, and note the conclusion that Tc-99 would decay very few times throughout a human lifetime. To show this, calculate the decay constant of Tc-99 and decay constant of Mo-99.

$$\lambda_{Tc99} = 3.29\times10^{-6}\text{ years}^{-1} = 3.75\times10^{-10}hour^{-1}$$

$$\lambda_{Mo99} = 92.15\text{ years}^{-1} = 0.25\ hour^{-1}$$

[You can also calculate the fraction of atoms that would remain after some period of time (i.e 80 years) to equally justify this solution.]

(c) Ruthenium is the only stable atom in the decay series. Determine the binding energy per nucleon of Ru-99.

$$\text{BE}/c^2 = \left[\left(44m_p + 55m_n + 44m_e\right) - m_{Ru}\right]$$

$$= \left[(1.65777\times10^{-25}kg) - (98.905393\times1.6605\times10^{-27}kg)\right]$$

$$\rightarrow \text{BE} = 1.54\times10^{-27}\times c^2 = 1.3584\times10^{-10}\text{J} = 865.1\text{ MeV}$$

$$\therefore \frac{\text{BE}}{A} = \frac{865.1\text{ MeV}}{99} = \mathbf{8.738\ MeV\ per\ nucleon}$$